

# 线性回归

# Linear Regression

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# 定义

## Definition

- ▶ 线性回归是一种监督式机器学习算法，用于建立因变量与一个或多个自变量之间的关系模型。该方法假设变量之间存在线性关系，其从已标记的数据集中学习，通过数据点学得最优线性函数，该函数可用于对新数据集进行预测。线性回归用于预测、趋势分析和数据洞察。

Linear regression is a type of supervised machine-learning algorithm, used to model the relationship between a dependent variable and one or more independent variables. The method assumes a linear relationship between the variables, by learning from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. Linear Regression is used for predictions, trend analysis, and data insights.

- ▶ 在线性回归中，所要学习的模型或最佳拟合线是最能准确表示自变量（输入）和因变量（输出）之间关系的直线。它是使实际数据点与模型预测值之间差异最小的直线。

In linear regression, the best-learned model or best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output). It is the line that minimizes the difference between the actual data points and the predicted values from the model.

# 定义

## Definition

- 在机器学习之前，线性回归的概念便已出现在概率统计中，其更一般的叫法是回归分析，目标是确定或估计与给定输入向量或自变量 $x$ 相关联的因变量 $y$ 。若 $y$ 是一个向量，则称之为多元回归。显然回归问题是一个连续模型。

Before machine learning, the concept of linear regression already existed in probability and statistics, where it is more generally known as regression analysis. Its goal is to identify or estimate the dependent variable  $y$  associated with a given input vector or independent variable  $x$ . If  $y$  is a vector, this is referred to as multiple regression. Clearly, a regression problem is a continuous model.

- 现在我们考虑一个房屋售卖的例子。假设房屋的售价 $y$ 与房屋面积 $x_1$ 与卧室的个数 $x_2$ 存在线性关系，如右表所示：

Now we consider a real estate example. Suppose that the price  $y$  of a property is linearly proportional to its area  $x_1$  and the number of bedrooms  $x_2$ , as shown in the right table:

| Area (sqm) | #Bedrooms | Price (CNY) |
|------------|-----------|-------------|
| 70         | 1         | 4,200,000   |
| 85         | 2         | 5,500,000   |
| 100        | 2         | 6,800,000   |
| 120        | 3         | 8,600,000   |
| 135        | 3         | 9,700,000   |
| 150        | 4         | 11,500,000  |

# 定义

## Definition

- ▶ 根据假设，我们有  $h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ ，其中， $\theta_i$  是参数（也称为权重），用于参数化从输入空间  $X$  到输出空间  $Y$  的线性变换。在不引起歧义的情况下， $h_{\theta}(x)$  中的下标  $\theta$  也可以省略，简记为  $h(x)$ 。为简化符号，令  $x_0 = 1$ ，于是  $h(x) = \sum_{i=0}^n \theta_i x_i = \theta^T x$ 。

According to the assumption, we have  $h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ . Here, the  $\theta_i$ 's are the parameters (also called weights) parameterizing the space of linear mapping from  $X$  to  $Y$ . When there is no risk of confusion, we will drop the  $\theta$  subscript in  $h_{\theta}(x)$ , and write it more simply as  $h(x)$ . To simplify our notation, we also introduce the convention of letting  $x_0 = 1$  (this is the intercept term), so that  $h(x) = \sum_{i=0}^n \theta_i x_i = \theta^T x$ .

- ▶ 实际上， $h(x)$  更一般的形式是  $h(x) = \sum_{i=0}^n \theta_i \phi_i(x) = \theta^T \cdot \phi(x)$ 。我们称  $\phi_i(x)$  为基函数或特征变换函数，其可能是非线性的，其中， $\phi_0(x) = 1$ 。最简单的情况下，我们有  $\phi_i(x) = x_i$ 。

Actually, a more general expression for  $h(x)$  is  $h(x) = \sum_{i=0}^n \theta_i \phi_i(x) = \theta^T \cdot \phi(x)$ . We call  $\phi_i(x)$  the base function or feature transform, which can be potential nonlinear, with  $\phi_0(x) = 1$ . In the simplest case, we use linear basis functions  $\phi_i(x) = x_i$ .

# 统计背景

## Statistical Context

- 实际上，在上面房屋售卖的例子中，除非一套房子已经卖出，否则我们并不知道最后的房价。而最后的房价究竟是多少，会有一些随机因素的影响。因此，即便房屋的售价 $y$ 与房屋面积 $x_1$ 与卧室的个数 $x_2$ 存在线性关系，其真实的模型也应该如下所示：

In fact, in the previous house price example above, unless a house has already been sold, we would not know the final house price. Actually, the final price is influenced by a number of random factors. Therefore, even if there is a linear relationship between the house price  $y$  and the house area  $x_1$  and the number of bedrooms  $x_2$ , the real model should be expressed as follows:

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon$$

- 注意，在上式中，从因变量的角度， $y$ 代表观念里的东西，但在实际中，我们没有办法处理存在于观念里的抽象的东西，因此一般就将其当作观测值处理。同时，在机器学习中，我们的目标是期望建一模型进行计算预测，因此，令 $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ 表示预测值，上式也写为 $y = \hat{y} + \varepsilon$ 。

In the above formula, note that from the perspective of the dependent variable,  $y$  represents a conceptual entity, but in practice, we have no way to deal with such abstract conceptual entities, so we generally treat it as the observed value. Meanwhile, in machine learning, our goal is to build a model, so we denote the predicted value  $\hat{y}$  as  $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ , and subsequently  $y = \hat{y} + \varepsilon$ .

# 内涵

## Intricacy

- ▶ 实际上，由于此部分的内容同时出现在机器学习与概率统计中，不同学科可能会用不同的术语，读者要根据具体的语境进行区分。

In fact, since regression appears in both machine learning and probability statistics, different disciplines may use different terminologies, and readers should distinguish them according to the concrete context.

- ▶ 注意，线性回归主要强调参数的线性性，因此下面几种形式皆是线性回归的例子：
  - ▶ 多元线性回归
$$Y = B_0 + B_1^T X + E$$
  - ▶ 多项式线性回归
$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$$
  - ▶ 线性回归
$$\ln y = \theta_0 + \theta_1 x + \theta_2 e^x + \varepsilon$$
- ▶ Note that linear regression emphasize the linearity in the parameters, so the following forms are all examples of linear regression:
  - ▶ Multiple Linear Regression:
$$Y = B_0 + B_1^T X + E$$
  - ▶ Polynomial Linear Regression:
$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$$
  - ▶ Linear Regression:
$$\ln y = \theta_0 + \theta_1 x + \theta_2 e^x + \varepsilon$$

# 求解

## Solution

- 我们从概率统计的角度考查线性回归的最简单的形式，即  $\hat{y} = \theta_0 + \theta_1 x$ ，假设训练样本集为  $\{x_i, y_i\}_{i=1}^n$ ，我们有以下推导过程：

We examine the simplest form of linear regression from the statistical perspective, namely,  $\hat{y} = \theta_0 + \theta_1 x$ . Assuming the training dataset is  $\{x_i, y_i\}_{i=1}^n$ , we have the following derivation:

$$\varepsilon = y - \hat{y} \sim \mathcal{N}(0, \sigma) \Rightarrow \sum_{i=1}^n \varepsilon_i = 0 \wedge \sigma \rightarrow 0 \Leftrightarrow \sum_{i=1}^n (\varepsilon_i - \bar{\varepsilon})^2 = \sum_{i=1}^n \varepsilon_i^2 \rightarrow 0$$

$$\sum_{i=1}^n \varepsilon_i^2 \rightarrow 0 \Rightarrow \sum_{i=1}^n (y_i - \hat{y}_i)^2 \rightarrow 0 \Rightarrow \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2 \rightarrow 0$$

$$\Rightarrow \begin{cases} \sum_{i=1}^n \frac{\partial (y_i - \theta_0 - \theta_1 x_i)^2}{\partial \theta_1} = 0 \Rightarrow \sum_{i=1}^n 2x_i (y_i - \theta_0 - \theta_1 x_i) = 0 \\ \sum_{i=1}^n \frac{\partial (y_i - \theta_0 - \theta_1 x_i)^2}{\partial \theta_2} = 0 \Rightarrow \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i) = 0 \end{cases}$$

# 求解

## Solution

$$\begin{aligned}\sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i) &= 0 \Rightarrow n\bar{y} - n\theta_0 - n\theta_1 \bar{x} = 0 \Rightarrow \theta_0 = \bar{y} - \theta_1 \bar{x} \\ \sum_{i=1}^n 2x_i(y_i - \theta_0 - \theta_1 x_i) &= 0 \Rightarrow \sum_{i=1}^n x_i y_i - \theta_0 \sum_{i=1}^n x_i - \theta_1 \sum_{i=1}^n x_i^2 = 0 \\ \sum_{i=1}^n x_i y_i - \theta_0 \sum_{i=1}^n x_i - \theta_1 \sum_{i=1}^n x_i^2 &= 0 \Rightarrow \sum_{i=1}^n x_i y_i - (\bar{y} - \theta_1 \bar{x}) \sum_{i=1}^n x_i - \theta_1 \sum_{i=1}^n x_i^2 = 0 \\ &\Rightarrow \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i + \theta_1 \bar{x} \sum_{i=1}^n x_i - \theta_1 \sum_{i=1}^n x_i^2 = 0 \\ &\Rightarrow \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i + \theta_1 \bar{x} \sum_{i=1}^n x_i - \theta_1 \sum_{i=1}^n x_i^2 = 0 \\ &\Rightarrow \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \bar{x} \bar{y} + \theta_1 \sum_{i=1}^n \bar{x}^2 - \theta_1 \sum_{i=1}^n x_i^2 = 0\end{aligned}$$

# 求解

## Solution

$$\begin{aligned}\because SS_x &\triangleq \text{var}(x) = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - 2x_i\bar{x} + \bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 = \sum_{i=1}^n x_i^2 - \sum_{i=1}^n \bar{x}^2 \\ SS_{xy} &\triangleq \text{var}(x, y) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y} \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \bar{y} - \sum_{i=1}^n \bar{x} y_i + \sum_{i=1}^n \bar{x} \bar{y} = \sum_{i=1}^n x_i y_i - n\bar{x} \bar{y} = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \bar{x} \bar{y} \\ \therefore \Rightarrow \theta_1 &= \frac{\sum_{i=1}^n x_i y_i - \sum_{i=1}^n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - \sum_{i=1}^n \bar{x}^2} = \frac{\text{cov}(x, y)}{\text{var}(x)}\end{aligned}$$

# 求解

## Solution

$$\begin{cases} \theta_0 = \bar{y} - \theta_1 \bar{x} \\ \theta_1 = \frac{\text{cov}(x, y)}{\text{var}(x)} \end{cases}$$

- ▶ 上面是基于概率统计方法的推导，从数学的角度，各个参数都有明确的解析形式。实际上，根据参数的取值，我们可以从如下两点进行直观的理解：
  - ▶ 函数过数据的均值点 $(\bar{x}, \bar{y})$ ；
  - ▶ 自变量与因变量的相关性从系数中得到了体现。

The above derivation is based on probability and statistics methods. From a mathematical perspective, each parameter has a clear analytical form. In practice, based on the values of the parameters, we can gain an intuitive understanding from the following two points:

- ▶ The function passes through the mean point of the data  $(\bar{x}, \bar{y})$ ;
- ▶ The correlation between the independent variable and the dependent variable is reflected in the coefficients.

# 机器学习背景

## ML Context

- ▶ 在机器学习中，特别当回归的形式比较复杂时，一般是采用梯度下降的方法。首先，我们按照机器学习的范式，来确定相关元素：

- ▶ 模型：显然，对于简单线性回归问题，我们的模型是  $y = b + wx$ ；
- ▶ 数据集：由于在模型中用  $y$  表示了预测值，那么真实值我们用  $\bar{y}$  来表示，即数据集为  $\{x_i, \bar{y}_i\}$ ；
- ▶ 学习方法：如上所说，我们将采用梯度下降方法，其具有普适性；
- ▶ 损失函数：我们希望预测值与真实值相差尽可能小，因此，
$$L(w) = \frac{1}{2} \sum_{i=1}^n (y_i - \bar{y}_i)^2。$$

- ▶ In machine learning, especially when the regression form is relatively complex, the gradient descent method is generally adopted. First, following the paradigm of machine learning, we define the relevant components:

- ▶ Model: For the simple linear regression problem, our model is clearly  $y = b + wx$ ;
- ▶ Dataset: Since  $y$  denotes the predicted value in the model, we use  $\bar{y}$  for the true value. Thus, the dataset is  $\{x_i, \bar{y}_i\}$ ;
- ▶ Learning method: As mentioned above, we will use gradient descent, which is universal;
- ▶ Loss function: We aim to minimize the difference between the predicted values and the true values. Therefore, the loss function is  $L(w) = \frac{1}{2} \sum_{i=1}^n (y_i - \bar{y}_i)^2。$

# 偏差

## Bias

- 如果我们对比以下两个式子，不难看出，其是等价的：

If we compare the two formulas in the below, which are apparently equivalent:

$$\begin{cases} h_{\theta}(x) = \theta_0 + \theta_1 x \\ y = b + wx \end{cases}$$

特别的， $\theta_0 = b$ 。但为什么有偏差这个名称呢？实际上，在对特定群体的认知中，我们往往只关心平均情况。我们期望从样本中估计的均值与群体均值是一致的，此时我们称为无偏估计。我们已经说过，上面两种形式，都过均值 $(\bar{x}, \bar{y})$ 。实际上， $\theta_0$ 或 $b$ 从数学上讲就是截距，其会控制直线离开均值点的程度，因此， $b$ 控制了是有偏估计还是无偏估计，偏的程度。

In particular,  $\theta_0 = b$ . But why is it called the bias term? In practice, when inferring properties of a specific population, we usually care only about the average. We expect the mean estimated from samples to be consistent with the population mean; this is called unbiased estimation. As mentioned earlier, both forms above pass through the mean point  $(\bar{x}, \bar{y})$ . Mathematically,  $\theta_0$  or  $b$  is also the intercept, which controls how far the line shifts away from the mean point. Therefore,  $b$  determines whether the estimator is biased or unbiased, as well as the degree of bias.

# 梯度下降

## Gradient

- 由于上面两种表达等价，为了使梯度下降有统一的形式，我们采用 $y = \theta_0 x_0 + \theta_1 x_1$ 的形式。然后，按照梯度下降算法，推导如下：

Due to the fact the above two expressions are equivalent, to have the gradient descent method expressed in an uniform way, we take the form  $y = \theta_0 x_0 + \theta_1 x_1$ . Then according to gradient descent algorithm, we deduce as in the following:

$$\begin{aligned}\frac{\partial}{\partial \theta_j} L(\theta) &= \frac{\partial}{\partial \theta_j} \left[ \frac{1}{2} \sum_{i=1}^n (y_i - \bar{y}_i)^2 \right] \\ &= \sum_{i=1}^n (y_i - \bar{y}_i) \frac{\partial y_i}{\partial \theta_j} \\ &= \sum_{i=1}^n (y_i - \bar{y}_i) \frac{\partial (\theta_0 x_0 + \theta_1 x_1)}{\partial \theta_j} = \sum_{i=1}^n (y_i - \bar{y}_i) x_i\end{aligned}$$

- 因此在机器学习背景下，我们更新参数的算法如下所示：

Therefore in the context of machine learning, the algorithm for updating the parameters is below:

Loop

for every  $j$

$$\theta_j := \theta_j - \alpha \sum_{i=1}^n (y_i - \bar{y}_i) x_i$$

Until converge

# 逻辑回归

## Logistic Regression

- 对于二分类问题，即  $y \in \{0, 1\}$ ，有时候我们又称其为逻辑回归。此时，我们采用如下形式，且令  $P(y = 1|x; \theta) = h_\theta(x)$ ， $P(y = 0|x; \theta) = 1 - h_\theta(x)$ ，将分类问题转为回归问题：

For binary classification problem, aka.,  $y \in \{0, 1\}$ , sometimes we also call it logistic regression. In this case, we the following form, and set  $P(y = 1|x; \theta) = h_\theta(x)$  and  $P(y = 0|x; \theta) = 1 - h_\theta(x)$ , subsequently convert the classification into regression:

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

- 实际上，我们可以将输出写成更加紧凑的形式， $p(y|x; \theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}$ ，当写成这种形式时，一个好处是我们通常假设样本是i.i.d.的，我们可以利用MLE直接写出损失函数。

In fact, we can write the output into a more compact form, that is  $p(y|x; \theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}$ . In this way, we can take for granted that samples are i.i.d., and by the virtue of MLE we can directly write the loss function:

$$L(\theta) = p(\vec{y}|X; \theta) = \prod_{i=1}^n p(y^{(i)}|x^{(i)}; \theta) = \prod_{i=1}^n (h_\theta(x^{(i)}))^{y^{(i)}} (1 - h_\theta(x^{(i)}))^{1-y^{(i)}}$$

# 逻辑回归

## Logistic Regression

- 在实际中，我们会进一步将损失写成等价的形式：

In practice, we will rewrite the loss function in an equivalent form:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n y^{(i)} h_{\theta}(x^{(i)}) + \sum_{i=1}^n (1 - y^{(i)}) (1 - h_{\theta}(x^{(i)}))$$

- 如果读者去进一步求导，会发现关于  $\ell(\theta)$  的梯度具有极简的形式，即  $\frac{\partial}{\partial \theta_j} L(\theta) = -\sum_{i=1}^n (y_i - \bar{y}_i) x_i$ 。由于MLE要求梯度上升，因此参数更新我们需按以下公式进行。但读者会发现他们最终具有一致的形式。

If the readers try to calculate the derivatives of  $\ell(\theta)$  with regard to  $\theta$ s, you will notice it has a succinct form, aka.,  $\frac{\partial}{\partial \theta_j} L(\theta) = -\sum_{i=1}^n (y_i - \bar{y}_i) x_i$ . However, since MLE requires gradient ascent, we need to change the parameter updating formula below. However the user will discover they have the same final form.

$$\theta_j := \theta_j - \alpha \sum_{i=1}^n (y_i - \bar{y}_i) x_i$$